
JOSHUA WARD

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Profile

Current Field Meteorologist at Washington State University's AgWeatherNet. Double minor in Mathematics and Astronomy. Active individual who enjoys hiking, mountain biking, and snowboarding when possible. Adaptable, prompt, and skilled at providing accurate information and boosting team performance with diligence and attention to detail.

Experience

NASA Student Research Airborne Program (SARP) Alumnus; Irvine, CA — Summer of 2022

- Aided in the operation of air quality instrumentation onboard NASA's DC-8 aircraft during a live mission
- Created, completed, and presented a research project on climate change and its effects on giant kelp across the west coast of North America

Great Smoky Mountain Rain Gauge Field Technician; Asheville, NC — 2021-2024

- Perform data retrieval, maintenance, and calibration of MetOne Model 380D and 382D Tipping Bucket Rain Gauges in remote locations throughout the Great Smoky Mountain National Park

Yerkes Observatory Intern, Full-Time; Williams Bay, WI — Summer of 2023

- Collaborated with astrophysicists to create an algorithm using python to match stars found in historical photographic astronomical plates to known stars in professional sky surveys
- Developed an algorithm using Python and LINUX environments that reduces astronomical images from Charged Coupled Devices (CCD's), converting them to usable scientific images
- Presented at the 243rd meeting of American Astronomical Society (AAS) on Jan. 9th, 2024

Field Meteorologist; Prosser, WA — Fall 2024 - Present

- Prepare, assemble, and install weather stations that collect the highest spatiotemporal weather data across Washington state while adhering to World Meteorological Organization (WMO) standards
 - Configure, install, and conduct preventative maintenance on various meteorological sensors, including Vaisala AQT560 ambient air quality sensors
 - Utilize the output of Quality Assurance (QA) and Quality Control (QC) algorithms to identify faulty or malfunctioning meteorological/ambient air quality sensors in real-time, using both raw and post-algorithm data
 - Lead the eastern Washington field meteorologist team in concurrently executing a range of projects, ensuring that project expenses are effectively managed
 - Serve as project lead for a user-focused AQI webpage redesign, ensuring timely and understandable air quality information for stakeholders
 - Provide decision support for stakeholders across Washington through detailed weather forecast discussions
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- Contribute expert commentary in Washington State Magazine podcast interviews, analyzing complex seasonal and regional weather phenomena across the U.S.
 - Analyze and interpret NWP models to create weekly forecasts for Washington and overnight low forecasts for agricultural stakeholders in central Washington

Education

University of North Carolina Asheville, Asheville, NC — Bachelors in Atmospheric Science, 2024

- Cumulative GPA—3.320
- Notable Courses - Air Pollution Formation and Impacts, Physics 1 & 2, Calculus 3, Differential Equations, Linear Algebra, Physical Climatology, Physical Oceanography, Meteorological Instrumentation, Atmospheric Thermodynamics
- Notable projects - Determining EPA Attainment Status Across North Carolina Counties; Precipitation Measurement Variations Between Different Rain Gauge Manufacturers

Southeastern Community College, Whiteville, NC — Associates in Science, 2019

- Associates Degree GPA—3.767
- Notable Courses—General Chemistry 1 & 2, General Biology 1, Calculus 1 & 2

Skills

- Deployment, installation, and maintenance of meteorological and ambient air quality sensors, including tipping bucket rain gauges, pyranometers, anemometers, radiation shields, aspirated temperature sensors, and air quality monitors
- Data analysis using Python libraries including Pandas, Matplotlib, Xarray, and NumPy for large datasets across multiple file formats
- Programming experience in CRBasic, Python, Jupyter Notebooks, Google Colab, Fortran, and Linux command-line environments
- Knowledge of National Ambient Air Quality Standards (NAAQS), including criteria pollutant thresholds and associated health impacts
- Understanding of photochemical processes involved in tropospheric ozone formation
- Construction and installation of weather-tower hardware using power tools and low-voltage DC systems
- Experience with Campbell Scientific LoggerNet and Device Configuration Utility (DevConfig) software
- Field team leadership and project coordination, including supervision of technicians, scheduling, training, and multi-site operations
- Tower Climber and Rescue Certification

Notable Societies/Programs

- Phi Theta Kappa Honor Society
 - Sigma Pi Sigma Honor Society
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